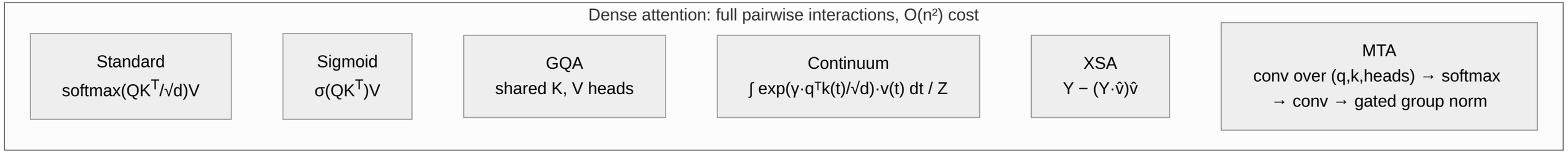
Dense attention: full pairwise interactions, O(n²) cost
Standard
softmax(QK^T/√d)V
Sigmoid
σ(QK^T)V
GQA
shared K, V heads
Continuum
∫ exp(y·q^Tk(t)/√d)·v(t) dt / Z
XSA
Y − (Y·v̂)v̂
MTA
conv over (q,k,heads) → softmax
→ conv → gated group norm